



Barret Hodgson Pakistan

Employee Off boarding Workflow Automation
Solution

Technical Documentation

For New Developers and IT Administrators

Version 1.0 | July 2026 | Barrett Hodgson IT Services

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1. Project Overview

1.1. Project Name

Barrett Hodgson Employee Off boarding Workflow Automation Solution

1.2. Project Purpose

Allow the IT administration to create an off boarding workflow case, and automatically route departmental clearance tasks through the configured workflow sequence.

1.3. Technology Stack

Table 1: Technology stack for the project:

Layer	Technology	Version/Implementation
Programming Language	Python	Python 3.9.13
Backend Framework	Flask	3.1.3
Database	SQLite	Built-in
Object-relational mapping	Flask-SQLALchemy	3.1.1
Database Migrations	Alembic, Flask-Migrate	1.16.5, and 4.1.0
Authentication	Flask-Login	0.6.3
Forms and validation	Flask-WTF	1.2.2
Frontend	HTML5, CSS, Vanilla JavaScript	N/A
Email notification	SMTP	Configurable through env variables, with a console mode available for local testing

2. System Architecture

2.1. High Level System Architecture

The application follows a simple layered web-application architecture:

2.1.1. Presentation Layer:

The presentation layer consists of server-rendered Jinja2 templates, CSS, and limited Vanilla JavaScript. Users can access the system through a web browser to create cases, complete departmental checklists, monitor progress, review final approval, view and print case details. Form validation messages and workflow status information are also displayed at this layer.

2.1.2. Application Layer:

The application layer is implemented through the Flask framework, and contains the system's routes, forms, authentication controls, and service modules. It manages case creation, workflow sequencing, checklist validation, secure task access, email notification processing, overdue monitoring, and final approval. The business logic is kept in separate dedicated services rather than being handled entirely inside the page routes.

2.1.3. Data Layer:

The application accesses the SQLite database through SQLAlchemy. The database stores users, cases, workflow tasks, checklist responses, access grants, notifications, and audit records. The application also communicates with the configured SMTP server to deliver departmental task links and workflow notifications.

2.2. Data Flow Diagram

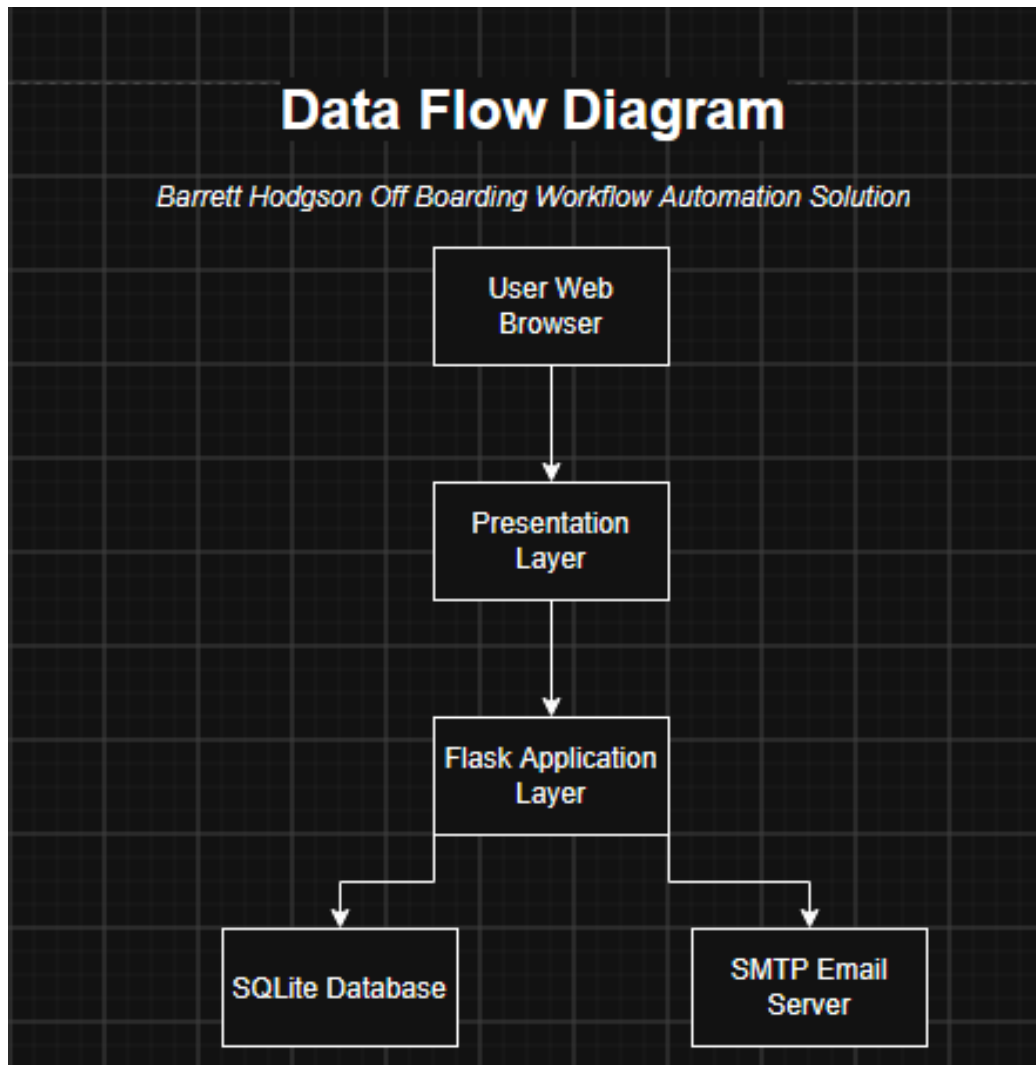


Figure 1 : Data flow diagram for the application

3. Functional Overview

The Off boarding Workflow Automation System manages employee clearance cases from initiation to final closure. It guides each case through the required departments, records checklist responses, sends task notifications, and provides authorised staff with visibility into current workflow status.

3.1. Case Creation

An authorized NOC user creates an offboarding case by entering the employee's details. The system validates the details, generates a unique case number, and initiates the workflow. The case creation action is also recorded in the audit history.

3.2. Departmental Clearance Workflow

Each case follows a predefined workflow currently consisting of NOC Access & Security Clearance, MIS Business and Application Clearance, Hardware and Asset Recovery, and Final Admin Approval. Only the current task is made available, with the next department being assigned its task after the previous one has been completed.

3.3. Checklist Completion

Each department only completes the checklist configured for its workflow. Each item on the checklist consists of three possible responses: "YES", "No", and "Not Applicable", along with a drop down consisting of that department's employees. For each checklist item, the user must select one of the three possible responses, along with selecting the name of the employee responsible for that response from the aforementioned employee name drop down. In case "No", or "Not Applicable" is selected, the user must provide a valid reason.

3.4. Email Notification and Secure Access

When a department task becomes active, the system sends an email containing the case reference, due date, and a secure task link. The recipient must use the secure task link to access the checklist. Notification attempts and delivery results are recorded by the system.

3.5. Case Monitoring and Printing

Authorized users (currently employees of the NOC department) can view the case register and open individual case records. The system displays the current workflow phase, completed tasks, checklist responses, and audit history. This allows pending or delayed clearances to be identified without the need of contacting each department individually. Authorized users can also print the relevant details of cases as well.

3.6. Overdue Task Handling

Each task is automatically assigned a due date based on the configured completion period. The system identifies active tasks that have passed their due date. Authorized users can then follow up with the concerned department and reissue a secure task link.

3.7. Final Approval and Case Closure

After all department's tasks have been completed, the case enters the Final Admin Approval Phase. The approver can examine all the department's checklist responses, as well as basic employee, and offboarding case information. After thoroughly reviewing each response, the admin can approve the case. Approval closes the case and records the final decision and completion time.

3.8. Basic End-to-End Workflow

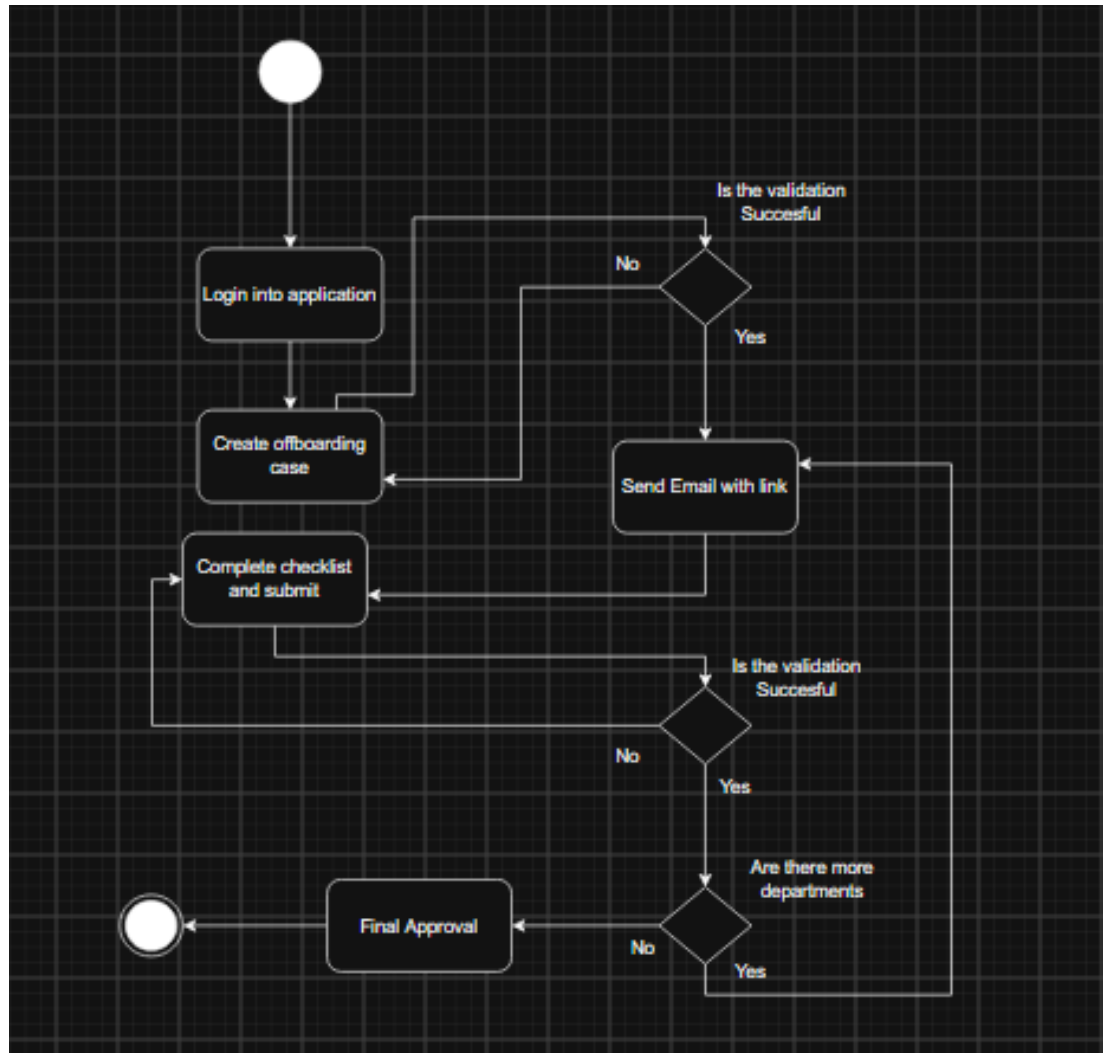


Figure 2: Activity diagram showcasing the basic flow of the application. Note: The activity diagram shows a highly abstracted version of the workflow

4. User Roles and Access

The system has three operational users: NOC, departmental users, and final approver. The NOC acts as the primary administrator of the off boarding process.

4.1. NOC User

The NOC user signs into the staff portal and manages the off boarding workflow, NOC user can create cases, view the case register, monitor progress, review notification results, print cases, and reissue departmental task links when necessary.

4.2. Departmental User

The departmental user can access their assigned checklist via a secure link generated by the system. They should not be given the login credentials to access the staff portal and should only be restrained to accessing their own departmental checklist.

4.3. Final Approver

The final approver reviews the completed checklists of each department and approves the case for closure.

5. User Interface and Operational Walkthrough

This section aims to present the main screens of the off boarding workflow automation system in the order in which they are normally used:

5.1. Staff Portal Sign-in

NOC users access the system through the staff sign-in page. The user enters their registered email address and password before being redirected to the dashboard. Invalid credentials prevent access to protected pages.

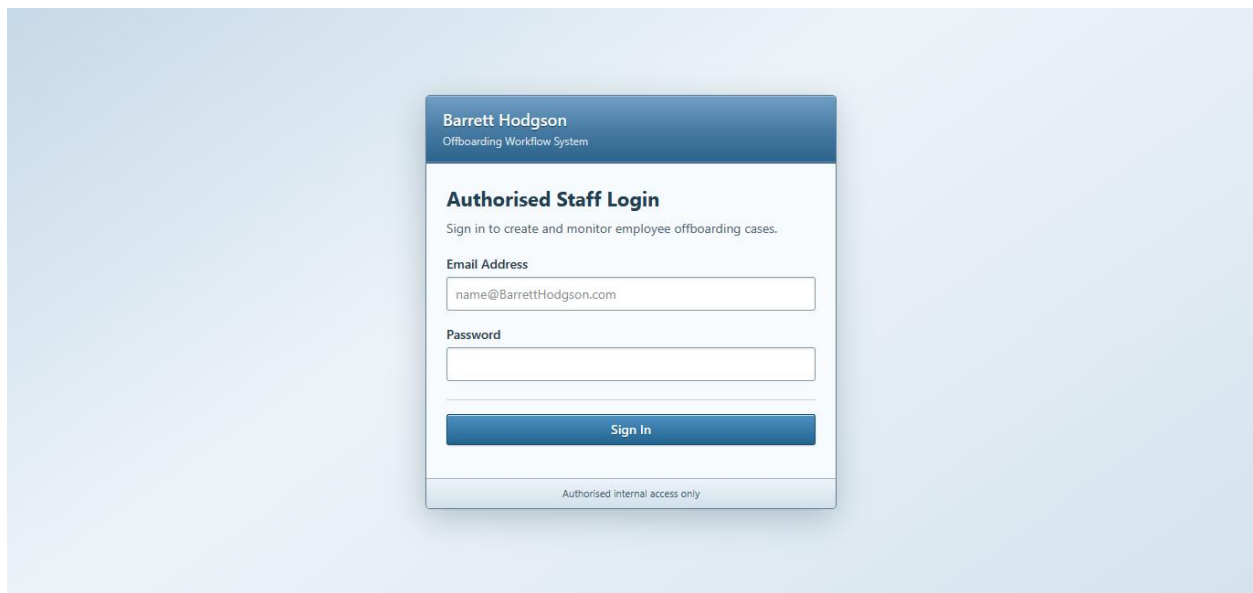


Figure 3: Staff Portal sign-in page

5.2. Dashboard

The dashboard allows the NOC user to view existing off boarding cases from one screen. It displays the case reference, employee information, current workflow stage, case status, and available actions. NOC users use this page to locate cases, view their information, and print them as well. The user also has the capability of searching, or filtering out the cases to only view the relevant ones.

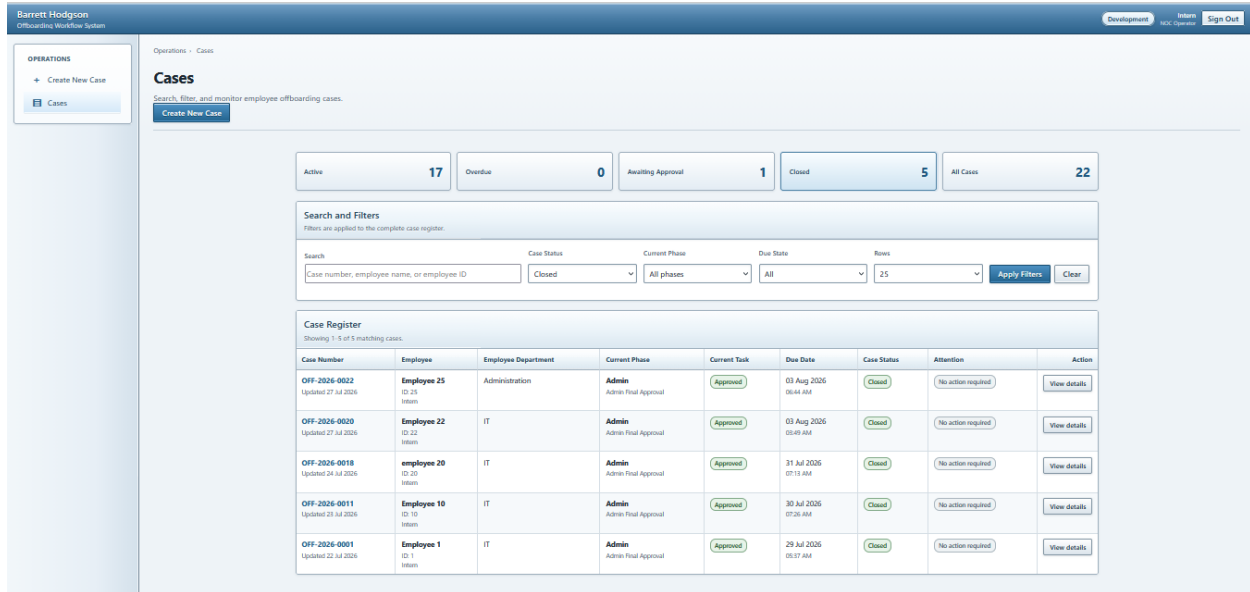


Figure 4: Case Dashboard accessible via Sign-in portal

5.3. Create Off boarding Case

NOC starts a new offboarding case by navigating to the “Create New Case” option located in the navigation bar. This option redirects them to a new off boarding case form, where the user must insert employee details, like employee name, department, last day of work, line manager, designation, and ID. Required fields are validated before the case is created. After which the workflow officially begins, and the user is redirected to a success page which confirms that the case has been successfully created along with relevant information about the off boarded employee.

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Development Intern NOC Operator Sign Out

OPERATIONS
+ Create New Case
Cases

Create a new offboarding case

Employee name
Employee

Employee designation
Intern

Employee id
30

Employee Department
Information Technology

Employee's last date of work
07 / 31 / 2026

Line manager/HOD
John Doe

Begin Offboarding Process

Figure 5: Off boarding case creation form

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Dashboard

Users

Sign Out

OPERATIONS

Creates New Case

Cases

Offboarding Case Created

The offboarding case and its initial workflow assignment have been created successfully

IN PROGRESS

Offboarding Case Information

Case Number	OFF-2026-0023	Employee Name	Employee
Employee ID	30	Employee Designation	Intern
Employee Department	Information Technology	Last Working Day	31 July 2026
Case Manager	John Doe	Created By	intern@barrethodgson.com
Created At	28 July 2026, 04:14 AM		

Initial Workflow Assignment

PENDING

Current Phase	NOC Access & Security Clearance	Assigned Department	NOC
Assigned Email	intern@barrethodgson.com	Task Status	PENDING
Assigned At	28 July 2026, 04:14 AM	Due At	04 August 2026, 04:14 AM

Task Notification

SENT

Notification Type	TASK_ASSIGNED	Recipient	intern@barrethodgson.com
Email Method	SMTP_DIRECT	Delivery Status	SENT
Assigned At	28 July 2026, 04:14 AM	Processed At	28 July 2026, 04:14 AM

Create Another Case

Figure 6: Case creation confirmation

5.4. Departmental Task access

When a departmental task becomes active, the assigned department receives an email with a secure task link. Opening the link takes the user directly to the departmental access page, without the need of signing into the application. The link is restricted to the assigned case and workflow.

Find Someone

Options

?

Replacement Offboarding Access Link — OFF-2026-0023

Tuesday, July 28, 2026 10:20 AM

Replacement Offboarding Access Link

A replacement secure link has been issued for this offboarding task. Any previous link and previously authorized browser session for this task are no longer valid.

Case Number	OFF-2026-0023
Employee	Employee
Employee ID	30
Phase	MIS Business Applications Clearance
Assigned Department	MIS
Due At	04 August 2026, 04:55 AM

[Open Assigned Task](#)

Security notice: This secure link is intended only for the assigned department. Do not forward or share it.

Figure 7: Figure 7: The assigned department receives the secure task link via Microsoft Outlook Web App. Note: Sensitive information has been redacted from this screenshot.

Departmental Clearance Assignment

Confirm that you want to open the assigned employee offboarding task.

Case Number	Employee
OFF-2026-0023	Employee
Responsible Department	Assigned Task
MIS	MIS Business Applications Clearance
Due Date	
04 August 2026, 04:55 AM	

Secure assignment link

This link provides access only to this workflow task. Do not forward or share it.

Open Assigned Task

Secure workflow task access

Figure 8: Departmental task access confirmation

5.5. Departmental Clearance Checklist

The departmental checklist displays the clearance items configured for the current workflow phase. For each item, the user selects a response along with the responsible employee. A reason must be given in case the user selects “No”, or “Not Applicable”.

[illegible]

Figure 9: Departmental checklist for phase "MIS Business Applications Clearance". Note: The page has been zoomed out.

5.6. Case Details, Progress Monitoring, Printing, and Reissuing Links

The case details page allows NOC user to review the current state of an offboarding case. It shows the employee information, workflow progress, completed departmental checklists, notification records, and audit activity. This page is also used to print relevant details of the selected case. In case a departmental secure link is inaccessible, the NOC user also has the capability to issue a new link.

Case Information

Case Number	OFF-2526-0023	Case Name	Case Name
Employee Name	Employee Name	Employee ID	Employee ID
Department	Department	Department Description	Department Description
Case Status	Case Status	Case Type	Case Type
Case Manager	Case Manager	Case Manager Email	Case Manager Email
Case Created	Case Created	Case Last Modified	Case Last Modified
Case Updated	Case Updated	Case Deleted	Case Deleted

Current Assignments

Assigned To	Assigned To	Assigned Department	Assigned Department
Assigned User	Assigned User	Assigned User Email	Assigned User Email
Assigned Date	Assigned Date	Assigned Date	Assigned Date
Assigned Time	Assigned Time	Assigned Time	Assigned Time

Workflow History

Step	Status	Assigned To	Date	Time	Assigned
Step 1: Case Assignment	Completed	Assigned User	2023-07-26	10:00:00	Assigned
Step 2: Case Review	In Progress	Assigned User	2023-07-26	10:00:00	Assigned
Step 3: Case Approval	Pending	Assigned User	2023-07-26	10:00:00	Assigned

Departmental Checklist Results

Item	Status	Assigned To	Date	Time	Assigned
Item 1: Case Assignment	Completed	Assigned User	2023-07-26	10:00:00	Assigned
Item 2: Case Review	In Progress	Assigned User	2023-07-26	10:00:00	Assigned
Item 3: Case Approval	Pending	Assigned User	2023-07-26	10:00:00	Assigned

Audit Activity

Activity	Date	User
Activity 1: Case Assignment	2023-07-26	Assigned User
Activity 2: Case Review	2023-07-26	Assigned User
Activity 3: Case Approval	2023-07-26	Assigned User

Figure 10: Complete details of a case, along with buttons to print case details, and reissue link.

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Operations - Cases - OFF-2026-0023 - Reissue Secure Link

Reissue Secure Link

OFF-2026-0023 - MIS

Cancel

Confirm Replacement Access
Review the current assignment before issuing a replacement secure link.

Previous access will be invalidated
Reissuing this link will revoke every previously issued secure link and invalidate any browser session currently authorized for this task.

Case Number OFF-2026-0023	Employee Employee
Employee ID 30	Current Phase MIS Business Applications Clearance
Department MIS	Recipient intern@barrethodgson.com
Task Status In Progress	Due Date 04 Aug 2026, 04:55 AM

Reason for reissuing the secure link
Record the operational reason for replacing the existing link. This reason will be retained in the case audit trail.

Example: The assigned department lost access to the original secure link.

Cancel Reissue and Send

Figure 11: NOC user can reissue a task link in case the previous link becomes inaccessible. The NOC user must use this feature with caution.

5.7. Final HOD Approval

After all departmental checklists have been submitted, the case becomes available for final review. The approver can inspect the case information, each departments responses, and approve the case. They also have the capability of leaving an optional remark.

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Final Offboarding Approval

Confirm that you want to open the assigned Administration approval task.

Case Number OFF-2026-0023	Employee Employee
Responsible Department Admin	Assigned Task Admin Final Approval
Due Date 04 August 2026, 06:07 AM	

Secure assignment link
This link provides access only to this workflow task. Do not forward or share it.

Open Final Approval

Secure workflow task access

Figure 12: Final off boarding form access confirmation

[illegible]

Figure 13: Final approval confirmation form

6. Workflow Processing Mechanism

6.1. Case Initialization

1. NOC submits the new offboarding case form.
2. Flask-WTF validates the submitted fields and CSRF token.
3. The selected employee department is verified against the active department records.
4. A new OffboardingCase record is created.
5. The workflow phases are retrieved in their configured sequence.
6. The first WorkflowTask is activated and assigned a due date.
7. A secure task-access link and notification record are created.
8. The case initialization is recorded in the audit history.

6.2. Departmental Checklist Submission

1. The departmental user opens the secure task link received by email.
2. The system validates the access token, expiry time, and task status.
3. The checklist configured for the current workflow phase is displayed.
4. Each response and responsible departmental employee is validated.
5. A reason is required when Not Applicable, or No is selected.
6. Valid responses are stored as ChecklistResponse records.
7. The current task is marked as completed.
8. The workflow progression process is then started.

6.3. Workflow Progression

1. Workflow tasks follow the sequence defined by the configured phases.
2. Only the current departmental task is available for completion.
3. Completion of the current task activates the next workflow phase.
4. A new secure link is generated for the next department.
5. The next assignment notification is created and sent.
6. After the last departmental task, the case moves to final approval.

Note: Workflow phases, checklist items, departments, and departmental employees must be configured before cases are processed.

6.4. Final Admin Approval & Closure

1. The final approver opens the completed case through the staff portal.
2. The system confirms that all required departmental tasks are complete.
3. The approver reviews the submitted checklist responses.
4. An approval decision is recorded.
5. Approval changes the case status to closed.
6. The decision and completion time are recorded in the audit history.
7. A rejected case remains available for further action according to the implemented workflow.

7. Secure Task Link Mechanism

7.1. Link Generation

1. A secure random token is generated when a departmental task becomes active.
2. The raw token is included in the task URL sent to the department.
3. Only a SHA-256 digest of the token is stored in the database.
4. The access grant is linked to a specific workflow task.
5. An expiry time is assigned according to the configured token lifetime.

7.2. Link Validation

When the task URL is opened, the system:

1. Extracts the token from the request
2. Calculates its SHA-256 digest
3. Searches for the corresponding *TaskAccessGrant* in the database (Refer to the [schema](#) section for seeing the corresponding table).
4. Verifies that the grant has not expired.
5. Checks that it has not been revoked.
6. Confirms that the associated task is still available.
7. allows access only when all checks succeed
8. Invalid, expired, or revoked links are rejected and do not expose the departmental checklist.

7.3. Link Reissue

NOC may reissue a task link when the original link is lost, expired, or considered compromised.

The following steps occur in such case:

1. Existing active grants for the task are revoked.
2. A new token and access grant are generated.
3. A replacement notification is sent to the configured department address.
4. The reissue action is recorded for audit purposes.

Security Note: Secure task URLs must not be exposed in screenshots, logs, or in any other place. The usable token is sent only through the departmental email, while the database only stores its hash.

8. Email Notification Mechanism

8.1. How notifications are sent

1. A notification record is created when a workflow task becomes active.
2. The application prepares the relevant content for email.
3. A secure task link is included where departmental action is required.
4. Delivery is attempted through the configured SMTP backend.
5. The notification status is updated after the attempt.
6. Delivery errors are stored for troubleshooting.
7. Console mode may be used during local development instead of sending an external email.

8.2. Notification Types

Table 2: Types of notifications,, and their purposes.

Notification	Purpose
Task assignment	Informs a department that its checklist is ready.
Replacement-link notification	Sends a newly issued secure task link.
Overdue escalation	Informs NOC that a task has passed its due date
Final-Approval Notification	Informs NOC that a task has passed its due date

8.3. Departmental Email Contents

A departmental assignment email normally includes:

1. Case reference.
2. Assigned department or workflow phase.
3. Employee Information
4. Due date
5. Secure task link.
6. Brief completion instructions.

8.4. Notification Tracking

Table 3:Types of notification statuses, and their purposes.

Notification Status	Meaning
Pending	Notification has been created but delivery is not yet confirmed
Sent	SMTP delivery completed successfully
Failed	Delivery was unsuccessful and an error was recorded

9. Security Implementation

9.1. Implemented Security Controls

Table 4:

Security Control	Implementation
Staff authentication	Flask-Login with email and password authentication
Password protection	Passwords stored as secure hashes rather than plaintext.
Role-based authorization	Protected portal routes restricted by operational role
Departmental access	Task-specific secure URLs instead of general portal access
Token protection	Only SHA-256 hashes are stored in database
Token expiry	Configurable expiry time for task-access links
Token revocation	Existing grants can be invalidated during link reissue or task submission.
CSRF protection	Flask-WTF CSRF validation on state-changing forms
Session Protection	Session cookies are marked HttpOnly to prevent JavaScript from accessing session cookies, and SameSite=Lax to limit cross-site transmission.

10. Database Schema and ERD

10.1. Database Overview

The Offboarding Workflow Automation System uses a relational database to store workflow configuration, employee's offboarding cases, departmental tasks, checklist responses, secure access records, email notifications, and audit events. The current prototype uses *SQLite* as the database management system, while *SQLAlchemy* is used as the Object-Relational Mapping framework. *SQLAlchemy* allows databases to be represented as Python model classes and enables the application to create, retrieve, update, and relate records without relying on manually writing SQL queries.

Flask-Migrate, which uses *Alembic* internally, is employed to manage the database schema changes. Migration files provide a controlled and traceable method for creating tables, adding columns, defining indexes, and modifying constraints as the application evolves.

The database consists of following main entities:

1. User
2. Department
3. DepartmentEmployee
4. WorkflowPhase
5. ChecklistItem
6. EmployeeDepartment
7. OffboardingCase
8. WorkflowTask
9. ChecklistResponse
10. TaskAccessGrant
11. AuditLog
12. EmailNotification

10.2. Entity-Relation Diagram (ERD)

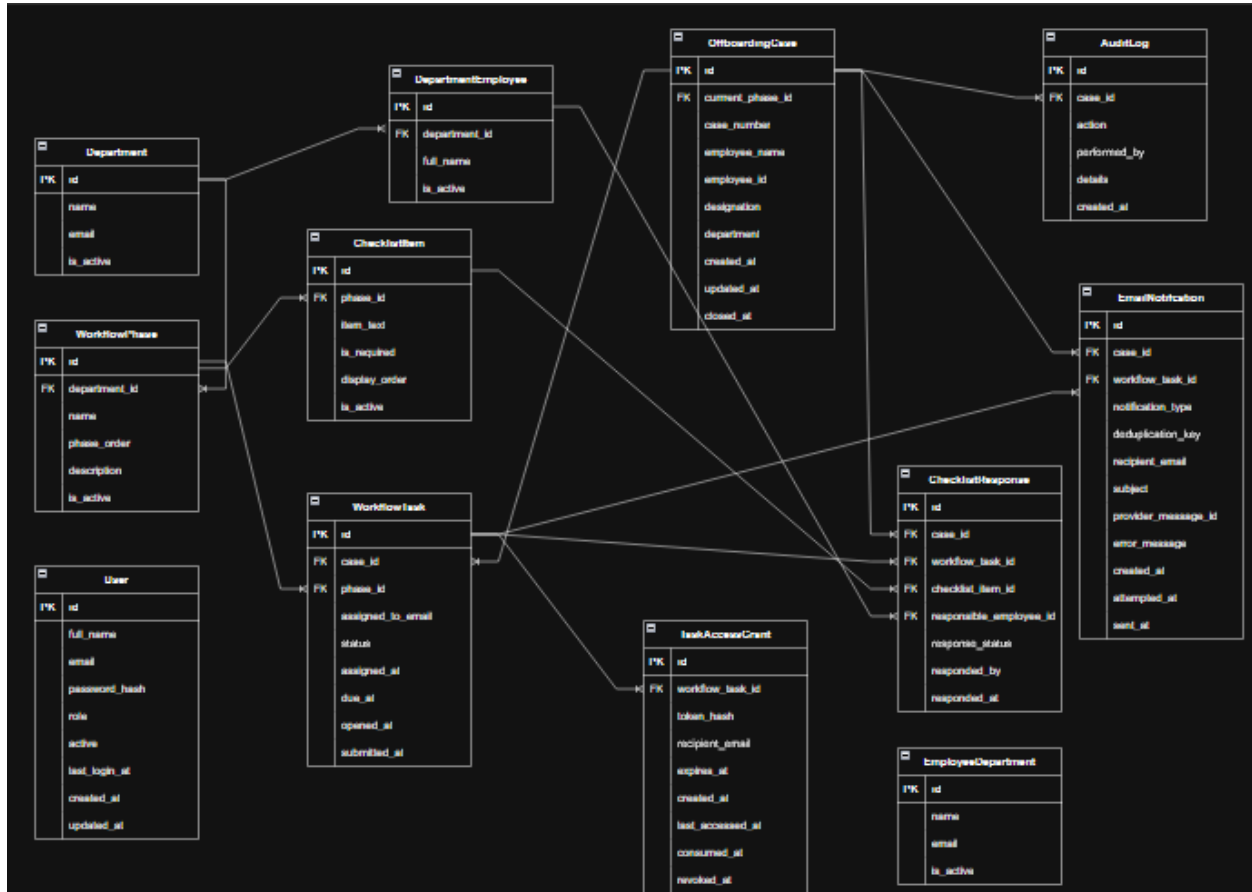


Figure 14: Entity-Relation Diagram (ERD) for the database schema of the application

10.3. Entity Descriptions

10.3.1. User

The User entity stores accounts permitted to access the authenticated operations portal. Each user is identified by a unique email address and has a stored password hash, full name, authorization role, active status, login timestamp, and record timestamps. Passwords are stored as hashes rather than plaintext values. The entity does not contain foreign keys to other database tables because portal authorization is role-based rather than directly linked to workflow departments.

10.3.2. Department

The Department entity represents departments that participate in the offboarding clearance workflow, such as NOC, MIS, Hardware, and Administration. Each department has a unique name, an email address used for workflow assignment and notification, and an active-status field. One department may contain multiple department employees and may be responsible for multiple workflow phases.

10.3.3. DepartmentEmployee

The DepartmentEmployee entity stores employees who may be selected as responsible for individual checklist responses. Each record belongs to one workflow department through `department_id` and contains a unique employee code, full name, and active status. The entity is connected to ChecklistResponse, allowing the system to record the specific departmental employee responsible for each clearance item.

10.3.4. WorkflowPhase

The WorkflowPhase entity defines the ordered stages through which an offboarding case progresses. Each phase belongs to one department and includes a name, unique slug, sequence number, optional description, active status, and an indicator specifying whether the phase represents final approval. A workflow phase may contain multiple checklist items and may be referenced by multiple workflow tasks.

10.3.5. ChecklistItem

The ChecklistItem entity defines an individual clearance requirement belonging to a workflow phase. Each checklist item contains the requirement text, an optional section heading, display order, required-status indicator, and active-status indicator. The `phase_id` foreign key determines the workflow phase in which the item appears. One checklist item may be referenced by multiple checklist responses across different offboarding cases.

10.3.6. EmployeeDepartment

The EmployeeDepartment entity stores the organizational departments of employees being offboarded, such as Finance, Sales, Human Resources, or Manufacturing. It is separate from the Department entity because it represents the employee's home department rather than a department responsible for performing workflow clearance. Each name is unique and may be deactivated through the `is_active` field. The table does not have a foreign-key relationship with OffboardingCase; the selected department name is stored in the case as text.

10.3.7. OffboardingCase

The OffboardingCase entity is the central operational record in the database. It stores the case number, employee information, designation, organizational department, final working date, line manager, case status, creator, timestamps, and current workflow phase. The `case_number` field is unique, while `current_phase_id` references the phase currently responsible for processing the case.

One offboarding case may contain multiple workflow tasks and audit-log records. The case also maintains `created_at`, `updated_at`, and `closed_at` timestamps to support lifecycle tracking.

10.3.8. WorkflowTask

The WorkflowTask entity represents the execution of one workflow phase for one offboarding case. Each task belongs to an OffboardingCase and a WorkflowPhase through the `case_id` and `phase_id` foreign keys. It records the assigned email address, task status, assignment time, due time, opening time, and submission time.

A unique constraint on `case_id` and `phase_id` ensures that the same workflow phase cannot generate more than one task for a particular case. One workflow task may contain multiple checklist responses, task-access grants, and email notifications.

10.3.9. ChecklistResponse

The `ChecklistResponse` entity stores the submitted result for one checklist item within a workflow task. Each response references an offboarding case, workflow task, checklist item, and responsible department employee. It also records the response status, optional reason, submitting identity, and submission timestamp.

A unique constraint on `workflow_task_id` and `checklist_item_id` prevents the same checklist item from receiving more than one stored response within the same task.

10.3.10. TaskAccessGrant

The `TaskAccessGrant` entity supports secure, task-specific access for departments that do not use permanent portal accounts. Each grant belongs to one workflow task and records the SHA-256 hash of the access token, intended recipient email, expiration time, creation time, access count, most recent access time, consumption time, and revocation time.

The raw token sent through email is not stored in the database. The `token_hash` field is unique, and multiple historical grants may exist for one task when links are replaced, revoked, or consumed.

10.3.11. AuditLog

The `AuditLog` entity stores a chronological history of significant actions associated with an offboarding case. Each record references one case and contains an action identifier, the responsible actor, optional event details, and the event timestamp. Examples of recorded events include task creation, task opening, checklist submission, secure-link activation, notification delivery, final approval, and case closure.

An index on `case_id` and `created_at` supports efficient retrieval of a case's audit history in chronological order.

10.3.12. EmailNotification

The `EmailNotification` entity records emails created and delivered during the workflow. Each notification is associated with both an offboarding case and a workflow task. It records the notification category, recipient email, subject, delivery status, optional deduplication key, provider message identifier, error details, creation time, attempt time, and successful sending time.

The entity supports assignment emails, replacement secure-link emails, and overdue escalation messages. Its indexes support queries by case and delivery status, as well as by workflow task and notification type.

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11. Operational Setup

11.1. Project Directory Structure

The source code is divided into folders according to operational responsibility. Route handlers, form definitions, business services, database models, interface templates, static assets, configuration, and migration files are maintained separately. This structure makes it easier to locate and modify a particular part of the system without placing all application logic in one file.

```
Offboarding-workflow-automation/  
|  
├─ app/  
|   ├── __init__.py  
|   ├── extension.py  
|   ├── models.py  
|   ├── comamnds.py  
|   ├── auth_commands.py  
|   ├── overdue_commands.py  
|   |  
|   ├── routes/  
|   |   ├── main_routes.py  
|   |   ├── auth_routes.py  
|   |   ├── case_routes.py  
|   |   ├── workflow_routes.py  
|   |   ├── task_access.py  
|   ├── forms/  
|   |   ├── auth_forms.py  
|   |   ├── offboarding_case.py  
|   |   ├── workflow_forms.py  
|   |   ├── case_action_forms.py  
|   ├── services/  
|   |   ├── workflow_service.py  
|   |   ├── checklist_service.py  
|   |   ├── approval_service.py  
|   |   ├── case_query_service.py
```

```
| | |——task_access_service.py
| | |——workflow_service.py
| | |——checklist_service.py
| | |——approval_service.py
| | |——case_query_service.py
| | |——task_access_reissue_service.py
| | |——notification_service.py
| | |——email_service.py
| |——templates/
| | |——base.html
| | |——layouts/
| | |——cases/
| | |——workflow/
| |——static/
| | |——css/
| | | |——app.css
| | | |——pages/
| | |——js/
| |
| |——migrations/
| | |——version
| | |——env.py
| | |——script.py.mako
|——config.py
|——run.py
|——requirements.txt
|——.gitignore
```

Figure 15: Directory structure of the project.

11.1.1. Application Directory

The app directory contains the main application source code. It is implemented as a Python package and contains the database models, route handlers, forms, service modules, templates, static files, and command-line utilities.

- **app/init.py**

The app/**init.py** file contains the application-factory function, `create_app()`. It creates the Flask application, loads the configuration, initializes the database, migration and login extensions, registers the application blueprints, and attaches the custom Flask command-line commands.

- **app/extension.py**

The extension.py file defines shared Flask extension objects, including the SQLAlchemy database object, Flask-Migrate integration, and Flask-Login manager.

These objects are defined separately from the application factory so they can be imported by models, routes, and services without creating circular imports. The objects are connected to the Flask application during execution of `create_app()`.

- **app/models.py**

The models.py file defines the SQLAlchemy model classes corresponding to the application's database tables. It contains the entities documented in [Chapter 10](#)

11.1.2. Routes Directory

The app/routes directory contains the HTTP route handlers. Routes receive browser requests, enforce the required access rules, process form submissions, call the relevant service functions, and return rendered templates or redirects.

The current route modules are:

Route module	Responsibility
main_routes.py	Handles the application's root and general navigation routes
auth_routes.py	Handles portal login, logout, and authenticated-session operations
case_routes.py	Handles case creation, case listing, case details, and case-level actions

workflow_routes.py	Handles departmental checklist tasks, task submission, and final approval
task_access_route.py	Handles activation and validation of secure task-access links

11.1.3. Forms Directory

Forms module	Responsibility
auth_forms.py	Login, logout, and authentication-related forms
offboarding_case.py	Employee offboarding case-creation form
workflow_forms.py	Departmental checklist and workflow-submission forms
case_action_forms.py	Case-level operational actions, including access-link reissue actions

11.1.4. Service Directory

The app/services directory contains the system's business and workflow-processing logic. Service functions are called by route handlers and command-line commands.

Service module	Responsibility
workflow_service.py	Creates workflow tasks and advances cases between phases
checklist_service.py	Validates and stores departmental checklist responses

workflow_forms.py	Departmental checklist and workflow-submission forms
case_action_forms.py	Case-level operational actions, including access-link reissue actions
approval_service.py	Processes final approval and case closure
case_query_service.py	Retrieves and filters cases for operational views
task_access_service.py	Creates, validates, consumes, and revokes secure task-access grants
task_access_reissue_service.py	Reissues task links and handles replacement-link operations
notification_service.py	Creates and manages workflow notification records
email_service.py	Sends or displays email messages through the configured backend

11.1.5. Commands Module

The application contains three command-related modules:

- **commands.py**
This file contains the data-seeding command used to create the initial workflow departments, phases, checklist items, department employees, and employee-department lookup records required by the application.
- **auth_commands.py**
This file contains the command used to create authenticated portal accounts. Portal-user creation is performed through a controlled command rather than through a publicly accessible registration page.
- **overdue_commands.py**
This file contains the process-overdue-tasks command. It identifies overdue workflow tasks and creates or retries the corresponding overdue notifications.